Repeated integrals and smooth endpoint data at every order

Research proof and partial Lean formalization
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Abstract

We prove the all-order endpoint-data conjecture of Burke, Haris and Madhavendra. The Stieltjes measure of a continuous nonconstant increasing highest derivative is atomless and nonzero. Its finitely many moments have a strictly positive smooth representing density equal to one near both endpoints. A direct separation argument proves this representation, and repeated integration supplies the required endpoint derivatives. The admissible endpoint-data set is an invertible linear image of an open convex smooth-moment image. The current Lean development verifies the source-to-measure bridge, the representation and witness assembly, with convexity and openness of that concrete image still explicit premises.

1 The source conjecture

For $n \geq 0$, let $W_n \subset \mathbb{R}^{n+1}$ consist of the vectors $b = (b_0, \dots, b_n)$ for which some $f \in C^n[0,1]$ satisfies

$$f^{(j)}(0) = 0, \qquad f^{(j)}(1) = b_j \quad (0 \leq j \leq n), \qquad f^{(n)} \text{ is nondecreasing and nonconstant.} \tag{1}$$

Endpoint derivatives are one-sided. Burke, Haris and Madhavendra [1], Conjecture 1.2, p. 2, ask whether the following statement (P_n) holds for every n. Their Theorem A proves it for $n \leq 3$.

Theorem 1. *For every integer $n \geq 0$, the set W_n is open. Every $b \in W_n$ has a witnessing $F \in C^\infty(\mathbb{R})$ such that*

$$F^{(n+1)}(x) > 0 \ (0 < x < 1), \qquad F^{(n+1)}(0) = F^{(n+1)}(1) = 1, \qquad F^{(j)}(0) = F^{(j)}(1) = 0 \ (j > n+1).$$

In fact $F^{(n+1)}$ can be chosen identically one on neighborhoods of both endpoints.

The source's sequel [2], Theorem 4.2, retains the corresponding all-order endpoint statement as a hypothesis. We prove (P_n) itself; we do not claim a separate discovery for its conditional applications or for the previously proved cases $n \leq 3$.

2 The smooth moment image

Put $p_n(t) = (1, t, \dots, t^n)$, and write $m(\mu) = \int p_n \, d\mu$ for the moment vector of a finite positive measure supported on $[0,1]$. Let

$$\mathcal{A}_n = \left\{ \int_0^1 p_n(t)\rho(t)\,dt : \begin{array}{l} \rho \in C^\infty(\mathbb{R}),\ \rho > 0 \text{ on } [0,1], \\ \rho = 1 \text{ near 0 and near 1} \end{array} \right\}. \tag{2}$$

All neighborhoods in this definition are neighborhoods in $\mathbb{R}$.

Lemma 2. *$\mathcal{A}_n$ is nonempty, convex and open, and $0 \in \overline{\mathcal{A}_n}$.*

Proof. The density one proves nonemptiness. Convex combinations preserve positivity and both endpoint germs. To prove openness at a vector represented by ρ, fix a nonnegative nonzero smooth function χ compactly supported in $(0,1)$. The matrix

$$G_{ij} = \int_0^1 \chi(t) t^{i+j}\,dt \qquad (0 \leq i, j \leq n)$$

is positive definite: its quadratic form at a nonzero coefficient vector c is $\int \chi(t)(\sum_i c_i t^i)^2 dt > 0$. Indeed χ is positive on some open interval and a nonzero polynomial has only finitely many roots. Hence G is invertible. For c in a sufficiently small neighborhood of zero, the densities

$$\rho_c(t) = \rho(t) + \chi(t) \sum_{i=0}^{n} c_i t^i$$

remain positive on $[0,1]$ and retain the endpoint germs. This uses the positive minimum of ρ and a uniform bound on the finite perturbation. Their moment vectors are $m(\rho\,dt) + Gc$, which contain a neighborhood.

For the closure assertion, take $0 < \delta < 1$ and smooth cutoffs $0 \le \chi_0, \chi_1 \le 1$, each equal to one near its endpoint, supported in the radius-δ ball around that endpoint. Choose $\delta < 1/2$ so each cutoff vanishes near the opposite endpoint. The density

$$\beta_\delta = \delta + (1-\delta)(\chi_0 + \chi_1)$$

belongs to the family in (2). Its mass on $[0,1]$ is at most $\delta + 4\delta = 5\delta$. Every coordinate of its moment vector lies between zero and 5δ, proving convergence to zero. □

Lemma 3. *Every finite nonzero atomless positive measure μ supported on $[0,1]$ has $m(\mu) \in \mathcal{A}_n$.*

Proof. Suppose $y = m(\mu) \notin \mathcal{A}_n$. Since $\mathcal{A}_n$ is open and convex, separation provides a continuous linear functional L such that

$$L(z) < L(y) \qquad (z \in \mathcal{A}_n). \tag{3}$$

The closure assertion gives $0 \le L(y)$. The functional is nonzero, since otherwise (3) contradicts nonemptiness.

We claim $L(p_n(t)) \le 0$ for every $t \in [0,1]$. If it is positive at an interior point, continuity supplies a nonnegative smooth bump ψ supported inside $(0,1)$ with

$$\int_0^1 L(p_n(t))\psi(t)\,dt > 0.$$

Every $1 + c\psi$, $c \ge 0$, is an allowed density in (2). Its L-moment tends to $+\infty$ as $c \to \infty$, contradicting (3). Continuity extends the claim to both endpoints.

The polynomial $q(t) = -L(p_n(t))$ is nonnegative on $[0,1]$ and nonzero: its coefficients are the negatives of L on the coordinate basis. Its zero set is finite, hence μ-null. Since μ has positive mass,

$$0 < \int q\,d\mu = -L(m(\mu)) = -L(y),$$

contradicting $0 \le L(y)$. This proves the lemma, including $n = 0$. □

3 Identification of the endpoint data

Let f witness $b \in W_n$ and set $g = f^{(n)}$. Then g is continuous, nondecreasing and nonconstant, and $g(0) = 0$. Extend it constantly outside $[0,1]$. Its Stieltjes measure $\mu = dg$ is finite, atomless, supported on $[0,1]$, and has positive mass $g(1)$. For every $0 \le j \le n$,

$$b_j = \frac{1}{(n-j)!}\int_0^1 (1-t)^{n-j}\,d\mu(t). \tag{4}$$

For $j = n$, this is $g(1) = \mu([0,1])$. For $k = n - j \ge 1$, Taylor's integral remainder with zero initial jets gives

$$b_j = \int_0^1 \frac{(1-x)^{k-1}}{(k-1)!} g(x)\,dx.$$

Substitute $g(x) = \mu([0,x])$ and interchange the integrals. The inner integral from t to one is $(1-t)^k/k!$, proving (4). No derivative of the original g is required.

Define the linear map $T_n : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ by

$$(T_n y)_j = \sum_{i=0}^{n} \frac{\binom{n-j}{i}(-1)^i}{(n-j)!}\, y_i, \qquad 0 \le j \le n, \tag{5}$$

with $\binom{k}{i} = 0$ for $i > k$. Before reversing the output coordinates, its matrix is lower triangular with diagonal $(-1)^k/k!$. Thus T_n is invertible, and the binomial theorem gives $b = T_n m(\mu)$. By Lemma 3, choose a density ρ as in (2) with the same moments. Set

$$F(x) = \frac{1}{n!}\int_0^x (x-t)^n \rho(t)\,dt \qquad (x \in \mathbb{R}). \tag{6}$$

Repeated differentiation gives $F^{(n+1)} = \rho$ and zero initial jets through n. Equations (4)–(5) give the terminal jets b_j. The endpoint germs of ρ give all higher endpoint conditions in Theorem 1.

Conversely, construction (6) for every density in (2) produces an element of W_n: its nth derivative is strictly increasing. Therefore

$$W_n = T_n(\mathcal{A}_n).$$

Lemma 2 and invertibility of T_n prove openness, completing Theorem 1.

4 Prior work and formal status

Smooth representation of interior truncated moment functionals is established in di Dio [3], Theorem 2.5 and Corollary 2.7 of arXiv v2; the latter is Corollary 3.1 in the published paper. These results belong to an earlier moment-theoretic framework, not new machinery here. Our direct proof uses standard separation, polynomial root finiteness and smooth bumps. The source-specific content is the atomless Stieltjes identification, the endpoint-one density requirement, and the all-order endpoint assembly. The preliminary source, mathematical, bounded prior-work and eligibility review passed with recorded limitations. No absolute priority claim is made.

The current pinned Lean 4.34.1/mathlib `d13f23b7` development has 103 declarations in 16 modules. A fresh isolated replay treats warnings as errors and checks every declaration's type and axioms; only `propext`, `Classical.choice` and `Quot.sound` occur. It formalizes the literal source hypotheses, Stieltjes measure and kernel identity, invertible map (5), all derivatives of the smooth witness, the endpoint baseline and the separation proof of Lemma 3. The final assembly

`RepeatedIntegrals.source_conjecture_of_moment_openness`

still takes convexity and openness of the *actual* set (2) as explicit premises. Their concrete Lean proofs remain to be integrated; the full source theorem is not yet a premise-free formal certificate. Final independent formal/manuscript review and packet replay are pending. This draft has discovery credit zero.

References

[1] M. R. Burke, M. Haris and Madhavendra, *Repeated integrals of increasing functions*, arXiv:2512.02151v1 (2025).

[2] M. R. Burke, *Comonotone approximation and interpolation by entire functions II*, arXiv:2512.23949v1 (2025).

[3] P. J. di Dio, *Absolutely continuous representing measures of moment sequences with an emphasis on the regularity of the densities*, arXiv:2411.03145v2 (2024). Published truncated-moment version: doi:10.1007/s00013-025-02175-1 (2025).